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An ultrathin, strong, flexible composite solid electrolyte for high-voltage lithium metal batteries

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The urgent need for safe, energy-dense electrochemical storage devices calls for high-voltage solid-state lithium metal batteries. However, state-of-the-art solid-state electrolytes are hindered by two critical issues: the narrow electrochemical window that prevents the pairing of a lithium metal anode with a high-voltage cathode, and the large thickness that leads to a huge internal resistance and a significant sacrifice in energy density. To simultaneously address the both issues, in this work we develop a poly(acrylonitrile) (PAN)-LiClO₄-boron nitride nanoflake (BNNF) composite electrolyte modified by a BNNF layer (PBCEB). The PAN-LiClO₄-BNNF composite can sustain an oxidation voltage of up to 4.5 V vs. Li/Li⁺ while the BNNF modifying layer prevents the PAN-LiClO₄-BNNF from the reduction reaction with the lithium metal anode. In the meantime, thanks to the BNNFs, the PAN-LiClO₄-BNNF possesses a strong tensile strength (16.0 MPa) and Young's modulus (563.7 MPa), which enables the PBCEB to be as thin as 13.5 μm (12.0 μm for the PAN-LiClO₄-BNNF and 1.5 μm for the BNNF modifying layer). As a result, a Li/PBCEB/LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ full battery delivers a high specific capacity of 173.6 mAh g⁻¹ at 0.2 C and achieves remarkable capacity retention of 68.1% after 350 cycles at 1 C. This work provides an effective approach to develop high-performance composite solid electrolytes for high-voltage lithium metal batteries.

Introduction

Since the commercialization by Sony back in the early 1990s, lithium-ion batteries (LIBs) have been widely applied ranging from portable devices to electric vehicles (EVs), which greatly revolutionize our lives.^{1–7} Unfortunately, conventional LIBs with graphite anodes are approaching their theoretical energy density limit (250 Wh kg⁻¹), which will no longer satisfy the unprecedented demand from modern products such as EVs that require a higher energy density (300 Wh kg⁻¹).^{8–11} Meanwhile, the use of toxic and flammable liquid electrolytes have caused many fire/explosion, posing a serious safety concern, that further limits their applications.^{12–15}

As an alternative, high-voltage solid-state lithium metal batteries have attracted intensive attention from both academia and industry due to their great potential to achieve both high energy density and enhanced safety.^{16–18} On the one hand, solid-state electrolytes (SSEs) with high shear modulus can inhibit the lithium dendrite growth, thus enabling the practical use of the lithium metal anode, which is considered to be the ultimate anode choice due to its ultrahigh specific capacity (3861 mAh g⁻¹) and the lowest electrochemical potential (-3.04 V vs. standard hydrogen electrode).^{19–26} Pairing lithium metal anodes with high-voltage cathodes, such as lithium nickel manganese cobalt oxide (LiNi_xMn_yCo_zO₂, NMC), is an effective approach to achieve high-energy-density batteries.^{27–29} On the other

hand, replacing liquid electrolytes with SSEs that are thermally stable promises safer batteries, which is a key indicator for widespread market adoption.^{12,30–35}

Despite the great promise, the advancement of high-voltage solid-state lithium metal batteries has been severely hindered by two critical issues of current SSEs. First, the electrochemical window of the extant SSEs is so narrow that they cannot be directly used to pair lithium metal anode with high-voltage cathodes.^{36–41} Second, the thicknesses of SSEs are generally too thick to ensure the satisfying mechanical properties, which leads to not only large internal resistance but also a huge sacrifice in both the gravimetric and volumetric energy density.^{42–44} Therefore, it is of an urgent need to develop advanced SSEs that are compatible with both electrodes and have high mechanical strength to achieve ultrathin thickness. To address the former issue, Zhou *et al.* proposed a double-layer polymer electrolyte, in which the poly(ethylene oxide) (PEO)-lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) layer provided stable contact with the lithium metal anode while the poly(N-methyl-malonic amide) (PMA)-LiTFSI layer allowed extraction of Li⁺ ions from a 4 V cathode without being oxidized.³⁸ Wang *et al.* prepared a three-tiered PEO-based solid polymer electrolyte, where the bilateral layers formed stable solid electrolyte interphase (SEI) and cathode electrolyte interphase (CEI) with the lithium metal anode and high-voltage LiCoO₂ cathode respectively.⁴⁵ Regarding the latter challenge, Wan *et al.* employed a robust polyimide (PI) film as the matrix which enabled a PI/PEO/LiTFSI polymer electrolyte with an ultralow thickness of 8.6 μm.⁴² Wu *et al.* fabricated an ultrathin (7.5 μm) polymer electrolyte by infiltrating the PEO-LiTFSI polymer electrolyte into the porous, high-strength polyethylene separator.⁴³ Although considerable progress has been made, SSEs that

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simultaneously meet the above two requirements are rarely reported.

In this work, we develop a poly(acrylonitrile) (PAN)-LiClO₄-boron nitride nanoflake (BNNF) composite electrolyte (PBCEB) modified by a BNNF layer to simultaneously address these challenges. As illustrated in Figure 1, the PBCEB comprises a BNNFs-doped PAN-LiClO₄ main body and a thin yet conformal BNNF modifying layer. The BNNF is chosen as the modifying and doping material because of its excellent electrochemical stability against both the lithium metal and high-voltage cathodes, and its high modulus to suppress lithium dendrite formation and enhance the mechanical strength of the composite electrolyte.^{46–49} The BNNFs of the modifying layer will protect the PAN-LiClO₄-BNNF from the reduction reaction with the lithium metal anode while the BNNFs in the PAN-LiClO₄-BNNF composite electrolyte can enhance the mechanical strength and ionic conductivity of the electrolyte. It is demonstrated that the PAN-LiClO₄-BNNF composite can sustain an oxidation voltage up to 4.5 V vs. Li/Li⁺, while the BNNF modifying layer can prevent the PAN-BNNF from reduction reaction with the lithium metal anode. Furthermore, the BNNFs endow the PAN-LiClO₄-BNNF with excellent tensile strength (16.0 MPa) and Young's modulus (563.7 MPa), enabling the ultrathin PBCEB of 13.5 μm (12.0 μm for the PAN-LiClO₄-BNNF and 1.5 μm for the BNNF layer), which minimizes the internal resistance and guarantees a high energy density of the solid-state battery. As a result, a Li/LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ high-voltage full battery can deliver a high capacity of 173.6 mAh g⁻¹ at 0.2 C and be stably cycled for over 350 cycles at 1 C with a capacity retention of 68.1%. The encouraging results prove that the novel bilayer structure proposed in this work represents a facile and effective strategy to fabricate ultrathin composite solid electrolytes for high-voltage lithium metal batteries.

Experimental

Preparation of the PAN-LiClO₄-BNNF composite electrolyte

The bulk BN powders were purchased from Sigma-Aldrich and the BNNFs were prepared using a method reported previously.⁵⁰ A certain amount of BNNFs was first ultrasonically dispersed in dimethylformamide (DMF). Then the PAN (*M*_w = 1,500,000, Sigma-Aldrich) and LiClO₄ (≥ 99.9%, Sigma-Aldrich) with a molar ratio of AN/Li⁺ = 5:1 were added into the slurry. After stirring for 5 h at 80 °C

to dissolve PAN and LiClO₄, the slurry was cast on a clean glass plate under ambient air conditions. Finally, the PAN-LiClO₄-BNNF composite electrolyte was obtained by vacuum drying at 65 °C for 24 h to remove the DMF solvent and then transferred and kept in a glovebox to avoid moisture contamination before use.

Preparation of the PBCEB

To achieve the PBCEB, a BNNF slurry was first prepared by dispersing/dissolving the BNNFs and polyvinylidene fluoride (PVDF, Shenzhen Kejing Star Technology Co., LTD.) with a mass ratio of 9:1 into the DMF solvent. The PVDF was used as a binder to maintain the integrity of the BNNF modifying layer during battery cycling due to its good electrochemical stability and binding strength.^{51,52} Then a spin coating method was employed to coat the BNNFs on the PAN-LiClO₄-BNNF composite electrolyte. The spin speed was 300 rpm for the first 3 seconds and 1500 rpm for the other 27 seconds. The PBCEB was finally obtained after vacuum drying for 24 h.

Material characterizations

The morphology observation was conducted by the JSM-7100 field emission scanning electron microscopy (FE-SEM) equipment operated with an acceleration voltage of 8 kV. The phase determination was performed by Philips high-resolution X-Ray Diffraction instruments with a Cu Kα radiation source operating at a current of 40 mA and a voltage of 40 kV. A JEM 2010F analytical transmission electron microscope was employed to obtain the TEM images of the BNNFs. The stress-strain curves of the PAN-LiClO₄-BNNF composite electrolyte and PAN-LiClO₄ polymer electrolyte was obtained by testing on the MTS Insight Electromechanical-2 kN Extended Length universal testing machine.

Electrochemical characterizations

The ionic conductivities of the electrolytes were determined by EIS tests using a potentiostat (EG&G Princeton, model 2273) with an amplitude of 10 mV in the frequency range of 7 MHz to 1 Hz. The electrolytes were sandwiched between stainless steel blocking electrodes and sealed in coin batteries to conduct the tests. The LSV mode with a sweeping rate of 1 mV s⁻¹ was used to measure the electrochemical stability of the electrolytes. The lithium metal and NCM811 were adopted as anode and cathode materials respectively

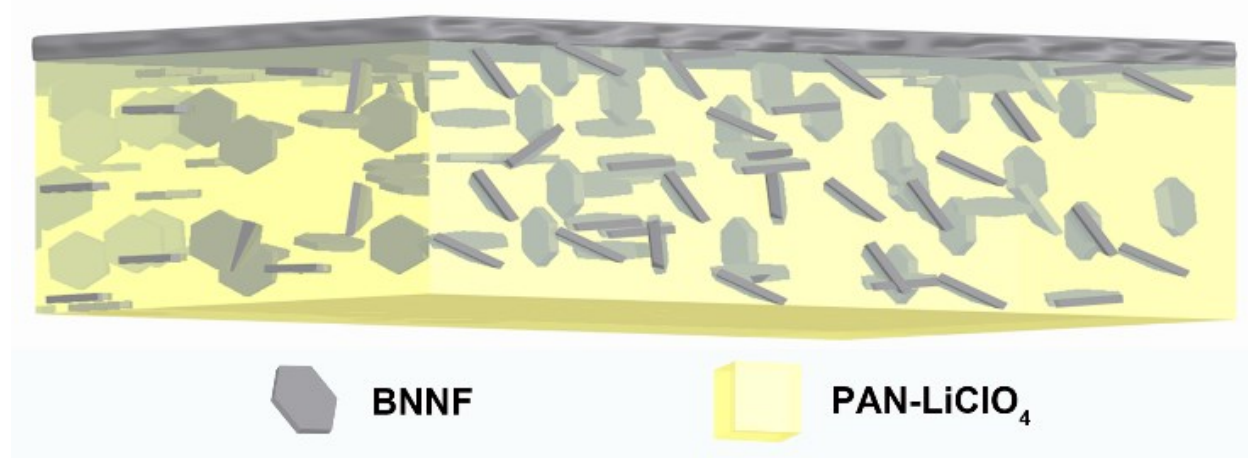


Figure 1. Schematic of the PBCEB.

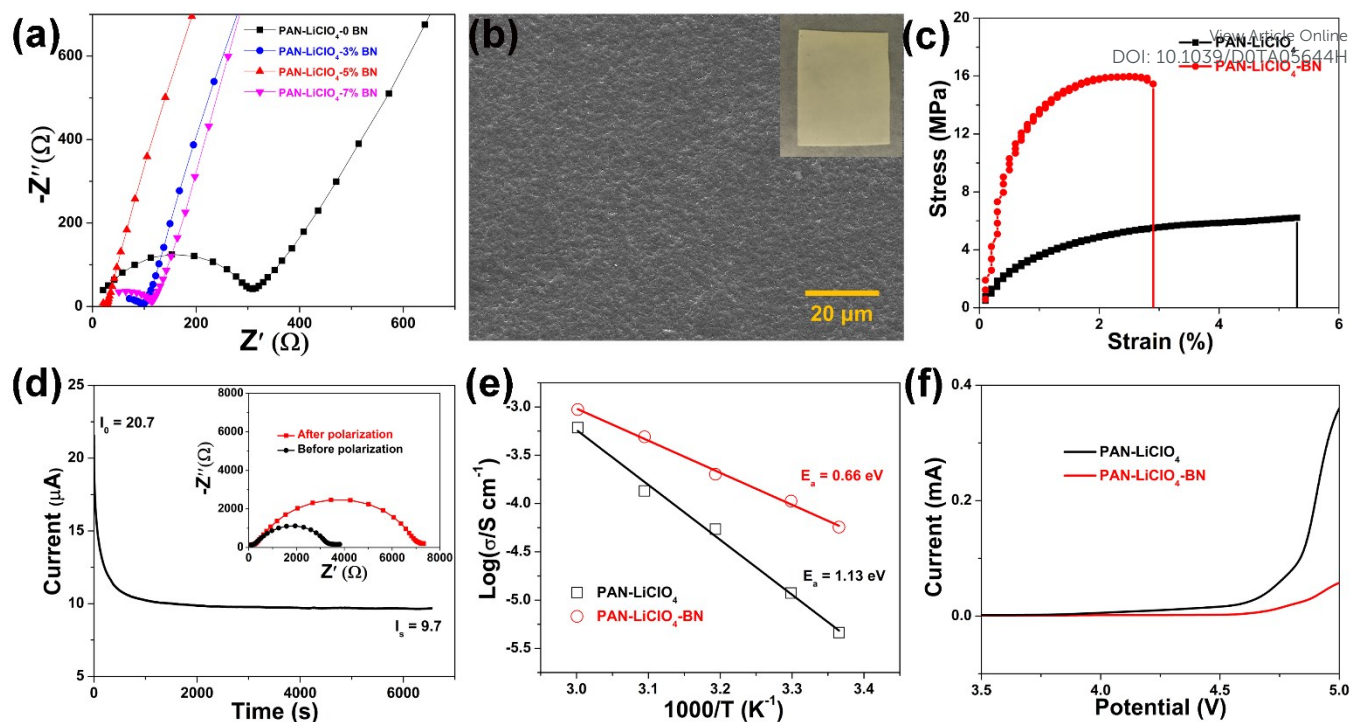


Figure 2. Characterizations of the PAN-LiClO₄-BNBF composite electrolyte. (a) Electrochemical impedance spectra (EIS) of the PAN-LiClO₄-BNBF composite electrolytes with 0 wt%, 3 wt%, 5 wt%, and 7 wt% BNBFs at room temperature; (b) SEM image of the surface of the PAN-LiClO₄-BNBF composite electrolyte. The inset is a photo of the composite electrolyte; (c) Stress-strain curves of the PAN-LiClO₄ and the composite electrolyte; (d) Chronoamperometry of the Li/PAN-LiClO₄-BNBF/Li symmetric cell with an applied voltage of 0.1 V. The inset is the impedance spectra of the symmetric cells before and after polarization; (e) Arrhenius plots of the PAN-LiClO₄ and the composite electrolyte; (f) LSV scans for the PAN-LiClO₄ and the composite electrolyte.

to assemble the high-voltage full batteries. The NCM811 cathode consisted of 70 wt% NCM811, 10 wt% PVDF, 10 wt% PAN, and 10 wt% carbon black. The mass loading of NCM811 was $3 \sim 5 \text{ mg cm}^{-2}$. The diameter of the NCM811 electrode used was 10 mm. During the battery assembly, a tiny amount of liquid electrolyte (1 M LiPF₆ in EC:DMC:DEC = 1:1:1 in volume ratio, 5 μL) was added on both electrodes to wet the interfaces and enable Li⁺ transport within the cathode, as demonstrated in previous works.^{53,54} All the batteries were assembled in a glovebox with both the H₂O and O₂ contents below 1.0 ppm. The battery cycling performance was performed on a Land 2001A battery testing system at room temperature.

Results and discussion

Characterizations of the PAN-LiClO₄-BNBF composite electrolyte

A simple solution casting method was used to prepare the PAN-LiClO₄-BNBF composite electrolyte. Figure S1 shows the transmission electron microscopy (TEM) image of BNBFs obtained by exfoliation of the hexagonal BN powders, from which it is seen that the diameter of the typical BNBF is $\sim 200 \text{ nm}$. It is widely acknowledged that doping ceramic fillers is an effective approach to enhance the ionic conductivity of the polymer electrolyte due to the formation of ion-ceramic interactions.^{55–58} To maximize the ionic conductivity, PAN-LiClO₄-BNBF composite electrolytes with different BNBF mass ratios were evaluated to determine the optimal BNBF content. As displayed in Figure 2a, the conductivities of samples with 0, 3, 5, and 7 wt% BNBFs are calculated to be 4.6×10^{-6} , 1.5×10^{-5} , 5.7×10^{-5} , and $1.2 \times 10^{-5} \text{ S cm}^{-1}$ respectively. The electrolyte with 5 wt% BNBFs

exhibits the highest ionic conductivity among all samples studied, which is an order higher than that of the pristine counterpart. Thus, the electrolyte with 5 wt% BNBF will be used for further investigations. The scanning electron microscopy (SEM) image in Figure 2b reveals that the composite electrolyte has a dense and uniform morphology, and the inset shows the pale color of the PAN-LiClO₄-BNBF composite electrolyte. The cross-sectional SEM images of the PAN-LiClO₄-BNBF are displayed in Figure S2. It is observed that BNBFs are evenly dispersed in the polymer matrix, indicating that the agglomeration of BNBFs is not very serious in the composite electrolyte. The XRD patterns of the PAN-LiClO₄ polymer electrolyte and the PAN-LiClO₄-BNBF composite electrolyte are shown in Figure S3, which confirms the mixture PAN and BN phases of the composite electrolyte. The TGA test was performed to clarify the amount of DMF solvent remaining in the composite electrolyte (displayed in Figure S4). The first weight loss below 200 °C can be attributed to evaporation of the DMF, thus its weight percentage is determined to be around 8 wt%.^{59,60} Figure 2c presents the stress-strain curves of these two electrolytes. It is found that compared with the pure PAN-LiClO₄ polymer electrolyte, both the tensile strength and Young's modulus of the PAN-LiClO₄-BNBF composite electrolyte are significantly enhanced, from 6.2 to 16.0 MPa and 212.3 to 563.7 MPa respectively. These dramatic improvements in mechanical properties can be attributed to the excellent intrinsic mechanical strength of the exfoliated BNBFs and their large surface area for interparticle interactions, which are consistent with previous works.^{46–48,57} Accordingly, the BNBFs doping enables the ultrathin composite electrolyte to achieve high mechanical strength, making the PAN-

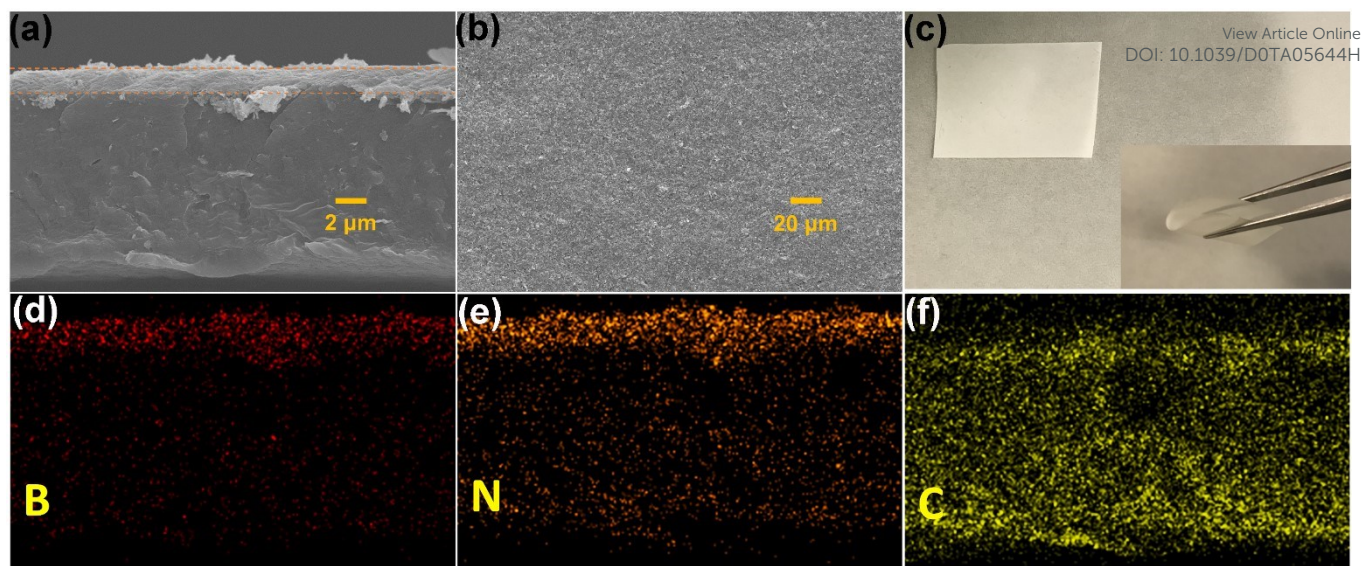


Figure 3. (a) Cross-sectional SEM image of the PBCEB; (b) Top SEM image of the BNNF layer of PBCEB; (c) A photo of the PBCEB showing the BNNF modified surface. The inset exhibits the excellent flexibility of the PBCEB; (d), (e), and (f) EDS mapping of the B, N, and C of the cross-sectional view in (a).

LiClO₄-BNNF composite electrolyte promising for practical battery application.^{42,43} Figure 2d shows the chronoamperometry of the Li/PAN-LiClO₄-BNNF/Li symmetric cell, and the inset is the electrochemical impedance spectroscopy (EIS) results before and after polarization, from which the transference number of the PAN-LiClO₄-BNNF composite electrolyte is calculated to be 0.39 based on the method reported by Evans *et al.*⁶¹, which is much higher than that ($t_{Li^+} = 0.20$) of the PAN-LiClO₄ polymer electrolyte (shown in Figure S5), indicating the dissociation of LiClO₄ is improved by introducing BNNFs. This is because B atoms in BN can interact with other Lewis bases due to its Lewis acidic characteristics, thereby facilitating the Li⁺ transport by attracting the anions in polymer electrolyte.^{48,57,62} Figure 2e displays the Arrhenius plots of these two electrolytes. The activation energies of the PAN-LiClO₄-BNNF and PAN-LiClO₄ are calculated to be 0.66 and 1.13 eV respectively according to the Arrhenius equation.⁶³ The greatly decreased activation energy of the PAN-LiClO₄-BNNF composite electrolyte verifies the effectiveness of BNNF doping in promoting Li⁺ transport. Figure 2f presents the linear sweep voltammetry (LSV) profiles of the PAN-LiClO₄ and the PAN-LiClO₄-BNNF composite electrolyte. It is observed that the oxidation tolerant voltage of the composite electrolyte is extended from 4.1 to 4.5 V. The enhancement in the electrochemical stability can be ascribed to the following two factors: i) the BN itself has high oxidation resistance; ii) the inorganic ceramic fillers in the composite electrolyte can remove the impurities (e.g. trace amount of H₂O) from the interfaces, thus improving the electrochemical stability.^{64–67} The BNNFs, which are reported to possess excellent adsorption ability originating from their large surface area, may trap water and other impurities, thus improving the electrochemical stability of the PAN-LiClO₄-BNNF composite electrolyte.^{68–71} Despite the excellent high-voltage stability of the PAN-LiClO₄-BNNF composite electrolyte, PAN-based electrolytes are reported to be unstable toward the lithium metal anode.^{72,73} As shown in Figure S6, an obvious reduction peak at around 1.5 V is observed in the LSV curve for the PAN-LiClO₄-BNNF scanning from 0 to 3 V, indicating its instability towards lithium

metal. That is to say, the PAN-LiClO₄-BNNF composite electrolyte itself cannot be directly applied in Li metal batteries, thus a protection layer is essential.

Material characterizations of the PBCEB

To address the above issue, a thin BNNF layer is modified on the surface to protect the PAN-LiClO₄-BNNF composite electrolyte via a facile spin coating method. From the cross-sectional SEM image (Figure 3a), it is seen that the prepared PBCEB displays an integrated bilayer structure, suggesting the extraordinary adhesion of the BNNF layer on the PAN-LiClO₄-BNNF. Figure S7 shows the structure of the BNNF modifying layer. It is seen that the BNNFs are well connected, indicating the good binding effect of the PVDF. The thicknesses of the modifying layer and composite electrolyte are 1.5 and 12.0 μm respectively, leading to a total thickness of only 13.5 μm of the PBCEB. As discussed above, the ultrathin thickness of the PBCEB is achieved owing to the superb mechanical strength of the PAN-LiClO₄-BNNF, which is the main section of the PBCEB. Figure 3d, e, and f show the energy dispersive spectroscopy (EDS) mapping results of the B, N, and C elements of the cross-sectional view in Figure 3a. It is seen that the B and N elements exist in both layers, while the C element is only detected in the PAN-LiClO₄-BNNF layer, confirming the integrated bilayer structure of the PBCEB. Figure 3b exhibits the uniform and conformal surface of the BNNF modifying layer on the PBCEB, which is beneficial to the suppression of lithium dendrite and protection of PAN-LiClO₄-BNNF from reduction with the lithium metal anode. The digital image in Figure 3c shows that the BNNF modified surface is white and free of visible defects, and the inset picture demonstrates the excellent flexibility of the PBCEB.

Electrochemical characterizations of the PBCEB

To verify the effectiveness of the BNNF modification in enhancing the electrochemical compatibility and stability of the PBCEB with the lithium metal anode, a PAN-LiClO₄-BNNF composite electrolyte with a BNNF protective layer on both sides (abbreviated as BNNF-PAN-LiClO₄-BNNF-BNNF) was prepared (Figure S8). Li/Li symmetric cells

using PAN-LiClO₄-BNNF and BNNF-PAN-LiClO₄-BNNF-BNNF were then assembled and evaluated. It should be noted that a tiny amount of liquid electrolyte (5 μ L) was added to the electrode/composite electrolyte interfaces during battery assembly to improve the interfacial contact and reduce the huge interfacial resistance, as demonstrated in previous works.^{53,54} Figure 4a and b show the voltage profiles of repeated lithium plating/stripping with a 2-hour period at the current densities of 0.15 mA cm⁻² (for the first 100 cycles) and 0.3 mA cm⁻² (for the following 100 cycles). It is seen that the two batteries display similar small overpotentials ($\sim$ 0.02 V at 0.15 mA cm⁻² and $\sim$ 0.04 V at 0.3 mA cm⁻²), suggesting the incorporation of BNNF modifying layers leads to a negligible increase in ohmic resistance of the battery. The EIS measurements of symmetric cells with the PAN-LiClO₄-BNNF and PBCEB were preformed, and results are shown in Figure S9. It can be seen that the symmetric cell using the PBCEB shows a slightly higher overall resistance than that using the PAN-LiClO₄-BNNF, further confirming that this BNNF modifying layer only adds limited resistance to the battery. The voltage curves of the symmetrical battery using BNNF-PAN-LiClO₄ composite electrolyte suffer from fluctuations at around 225 h and a short circuit after 360 h. The poor performance of the

Li/PAN-LiClO₄-BNNF/Li cell can be ascribed to the deteriorating interfaces between the electrolyte and lithium metal anode as a result of the adverse reactions. This is further confirmed by the dramatically changed PAN-LiClO₄-BNNF electrolyte and lithium metal plate after disassembling the failed cell. As exhibited by the inset of Figure 4a, the PAN-LiClO₄-BNNF electrolyte turns to deep yellow after failure, and both the electrolyte and lithium metal are covered by gray contaminants, which should be the products of side reactions. By contrast, the symmetric battery with the BNNF-PAN-LiClO₄-BNNF-BNNF exhibits stable and flat voltage profiles during cycling as shown in Figure 4b, demonstrating the superb interfacial compatibility and stability of the electrolyte in the presence of the BNNF modifying layer. As shown in the insets of Figure 4b, no obvious surface degradation can be detected from both the BNNF-PAN-LiClO₄-BNNF-BNNF electrolyte and lithium anode after cycling, further proving the effectiveness of the BNNF modifying layer.

To evaluate the feasibility of the PBCEB in high-voltage lithium metal batteries, Li/PBCEB/LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ (NCM 811) full batteries were assembled and compared to the Li/PAN-LiClO₄-BNNF/NCM811 batteries. Figure 5a shows the charge and discharge profiles of the Li/PBCEB/NCM811 full battery under different rates at

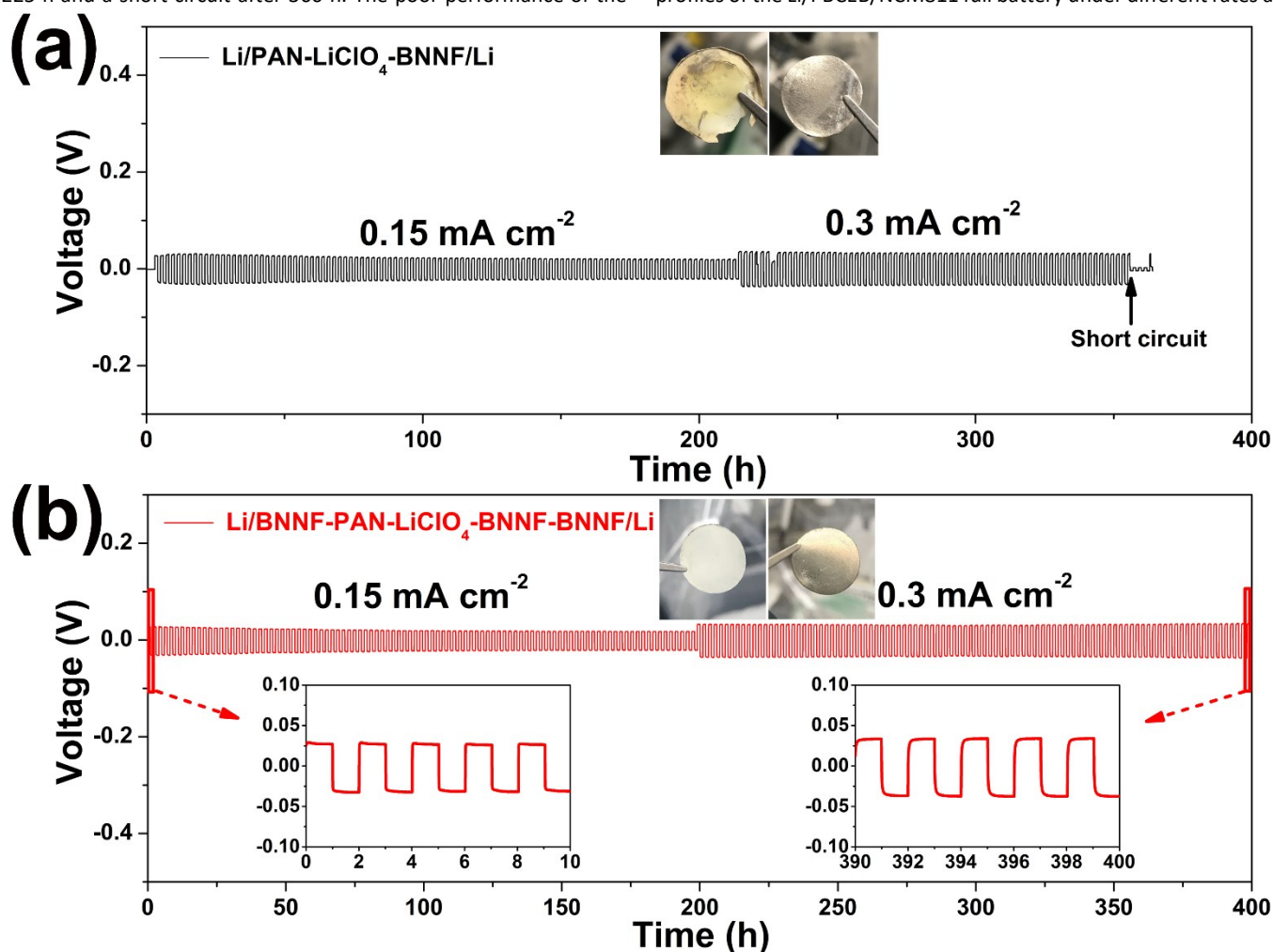


Figure 4. (a) and (b) Voltage profiles of long-term lithium plating/stripping cycles with a 2-hour period in Li/PAN-LiClO₄-BNNF/Li and Li/BNNF-PAN-LiClO₄-BNNF-BNNF/Li symmetric cells at the current density of 0.15 mA cm⁻² (for the first 100 cycles) and 0.3 mA cm⁻² (for the following 100 cycles) at room temperature. The insets in (a) and (b) show the PAN-LiClO₄-BNNF and BNNF-PAN-LiClO₄-BNNF-BNNF electrolytes and lithium metal anodes after cycles.

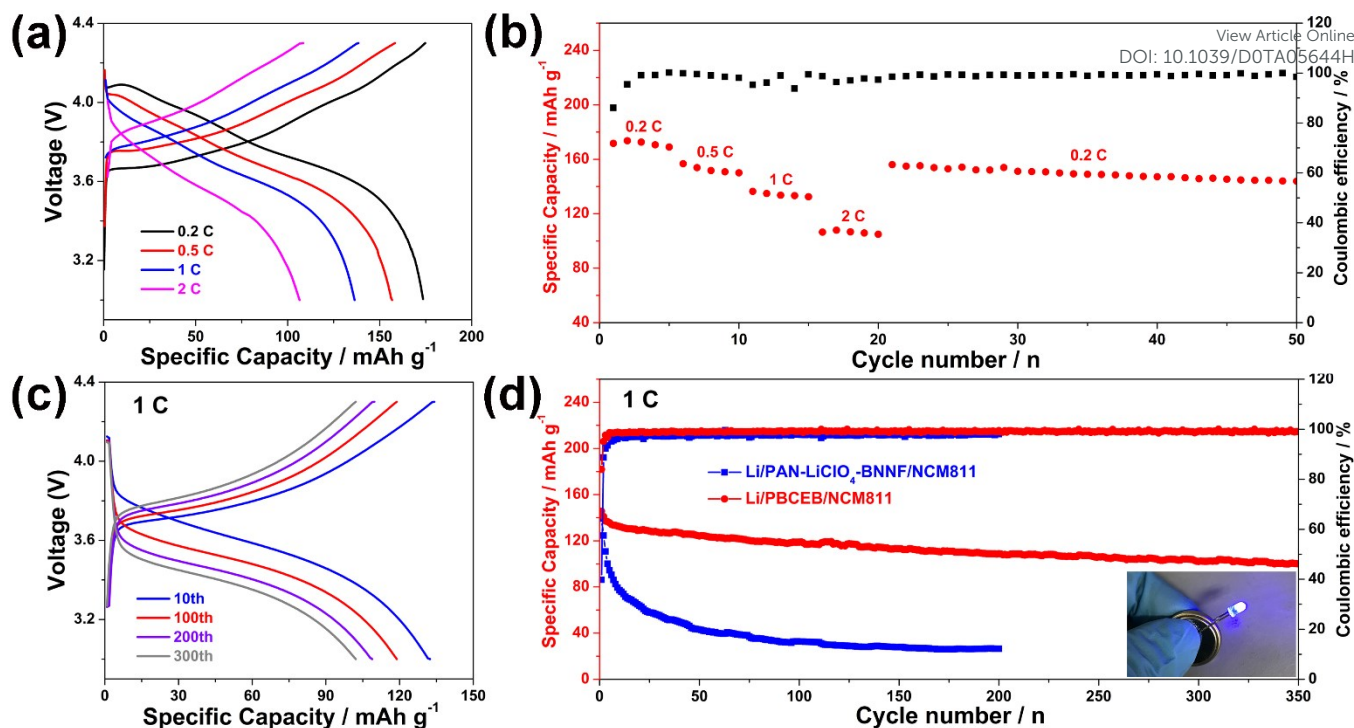


Figure 5. (a) Charge/discharge profiles of the Li/PBCEB/NCM811 full battery under different rates at room temperature; (b) C-rate performance of the Li/PBCEB/NCM811 full battery at room temperature; (c) Representative charge/discharge profiles of the Li/PBCEB/NCM811 full battery at 1 C at room temperature; (d) Long-term cycling performance of the Li/PAN-LiClO₄-BNNF/NCM811 and the Li/PBCEB/NCM811 full batteries at 1 C at room temperature. The inset shows a LED powered by the Li/PBCEB/NCM811 battery.

room temperature. The smooth curves without any fluctuations in the voltage range of 0 ~ 4.3 V vs. Li/Li⁺ confirm the compatibility of the PBCEB with both the lithium metal anode and high-voltage cathode. The C-rate performance of the Li/PBCEB/NCM811 full battery is displayed in Figure 5b. Stable coulombic efficiencies (~ 100%) and high specific capacities of around 173, 156, 135, and 106 mAh g⁻¹ are obtained at 0.2, 0.5, 1, and 2 C respectively. When swiftly changing the rate from 2 to 0.2 C, the capacity recovers to 156 mAh g⁻¹ after cycling at different rates, demonstrating the excellent rate capability and stability of the battery. Figure 5c exhibits the charge and discharge profiles of the Li/PBCEB/NCM811 full battery at the 10th, 100th, 200th, and 300th cycle at 1 C, which proves that the PBCEB can endure long-term battery cycling. Moreover, as displayed in Figure 5d, the Li/PBCEB/NCM811 full battery is capable of being stably cycled for over 350 cycles at 1 C with a capacity retention of 68.1%, whereas the battery using PAN-LiClO₄-BNNF suffers from relatively lower coulombic efficiencies and rapid capacity decay with a capacity retention of only 19.0% after 200 cycles due to the poor interface stability. The inset in Figure 5d displays a LED powered by the Li/PBCEB/NCM811 battery, which shows the potential of the fabricated PBCEB to be used for real applications. These encouraging results demonstrate that PBCEB holds great promise for high-performance high-voltage solid-state lithium metal batteries.

Conclusions

In summary, an ultrathin BNNF layer-modified PAN-LiClO₄-BNNF composite electrolyte is developed for high-voltage lithium metal batteries. The PAN-LiClO₄-BNNF composite with high oxidation resistance is designed to enable the high-voltage cathode, while the

BNNF modifying layer is constructed to prevent the side reactions between the lithium metal and PAN-LiClO₄-BNNF. Meanwhile, the BNNFs endow the PAN-LiClO₄-BNNF composite with excellent mechanical strength that enables the PBCEB to be as thin as 13.5 μm, which is beneficial for minimizing the internal resistance and realizing high energy density. As a result, a Li/PBCEB/NCM811 full battery is capable of delivering high specific capacities of 173.6 mAh g⁻¹ at 0.2 C and 135.6 mAh g⁻¹ at 1 C. Additionally, the battery can be stably cycled for over 350 cycles at 1 C with a capacity retention of 68.1%. This work offers a new avenue to fabricate ultrathin and high-performance composite solid electrolytes, thereby paving the way for the development of safe, high-voltage lithium metal batteries with high energy density.

Conflicts of interest

A Chinese patent (202010196325.7) has been filed.

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